Telescoping sums

High school · sequences · Worked solutions · 10

Exercise 1. Find $\sum[\frac{1}{k} - \frac{1}{k+1}]$ from $k = 3$ to 10.

Solution. Interior terms cancel: $\frac{1}{3} - \frac{1}{11} = \frac{8}{33}$.

Exercise 2. Express $\frac{1}{k(k+2)}$ as a difference of two fractions.

Solution. $\frac{1}{k(k+2)} = (\frac{1}{2})[\frac{1}{k} - \frac{1}{k+2}]$.

Exercise 3. Find $\sum(2k+1)$ from $k = 0$ to n using $(k+1)^2 - k^2$.

Solution. $2k + 1 = (k+1)^2 - k^2$. The sum is $(n+1)^2 - 0^2 = (n+1)^2$.

Exercise 4. If $S_n = \frac{n}{n+1}$, find the least integer $n \geq 1$ with $1 - S_n < 0.01$.

Solution. $\frac{1}{n+1} < \frac{1}{100}$ means $n + 1 > 100$, hence $n \geq 100$. The minimum is 100.

Problem · Investigate, justify, conclude. Let $T_n = \sum \frac{1}{k(k+2)}$ for k from 1 to n. Derive an exact formula and find its limit. Find the least n such that the gap to the limit is strictly less than 0.01.

Solution. After decomposition, all except the first two and last two terms cancel: $T_n = \frac{3}{4} - \frac{1}{2(n+1)} - \frac{1}{2(n+2)}$. The limit is $\frac{3}{4}$. The gap decreases with n. At $n = 98$ it is $\frac{1}{198} + \frac{1}{200} > 0.01$; at $n = 99$ it is $\frac{1}{200} + \frac{1}{202} < 0.01$. Therefore the minimum is 99.